Assignment

1 About the Team and Assignment

1.1 Class

IT2151-01

1.2 Focus Area

Smart Home Energy Monitor

1.3 Team Lead

Zwe Htet Kyaw

1.4 Members

1) Zwe Htet Kyaw (243402M)
2) Chan Ken Yew (243204Q)
3) Subramanian Tharun Ranjan (241798F)
4) Choon Lin (240602F)
5) Tan Pekyu (240616H)

2 Product Vision Statement

2.1 Product Vision Statement

The ***Smart Home Energy Monitor*** is an IoT-powered solution designed for modern **homeowners** seeking **better visibility and control over their energy consumption**. In an age of rising utility costs and growing concern for environmental sustainability, many users still rely on outdated billing systems that provide limited insights and delayed feedback.

Our solution addresses this gap by offering **real-time energy tracking**, **intelligent usage analytics**, and **proactive notifications**, empowering users to make informed decisions that reduce waste and save costs. At the same time, the system supports **technical administrators** who **oversee infrastructure performance and ensure smooth operation** across multiple environments.

By aligning with global trends in smart living and energy efficiency, the Smart Home Energy Monitor not only enhances user experience but also contributes meaningfully to **sustainable energy practices and responsible consumption**.

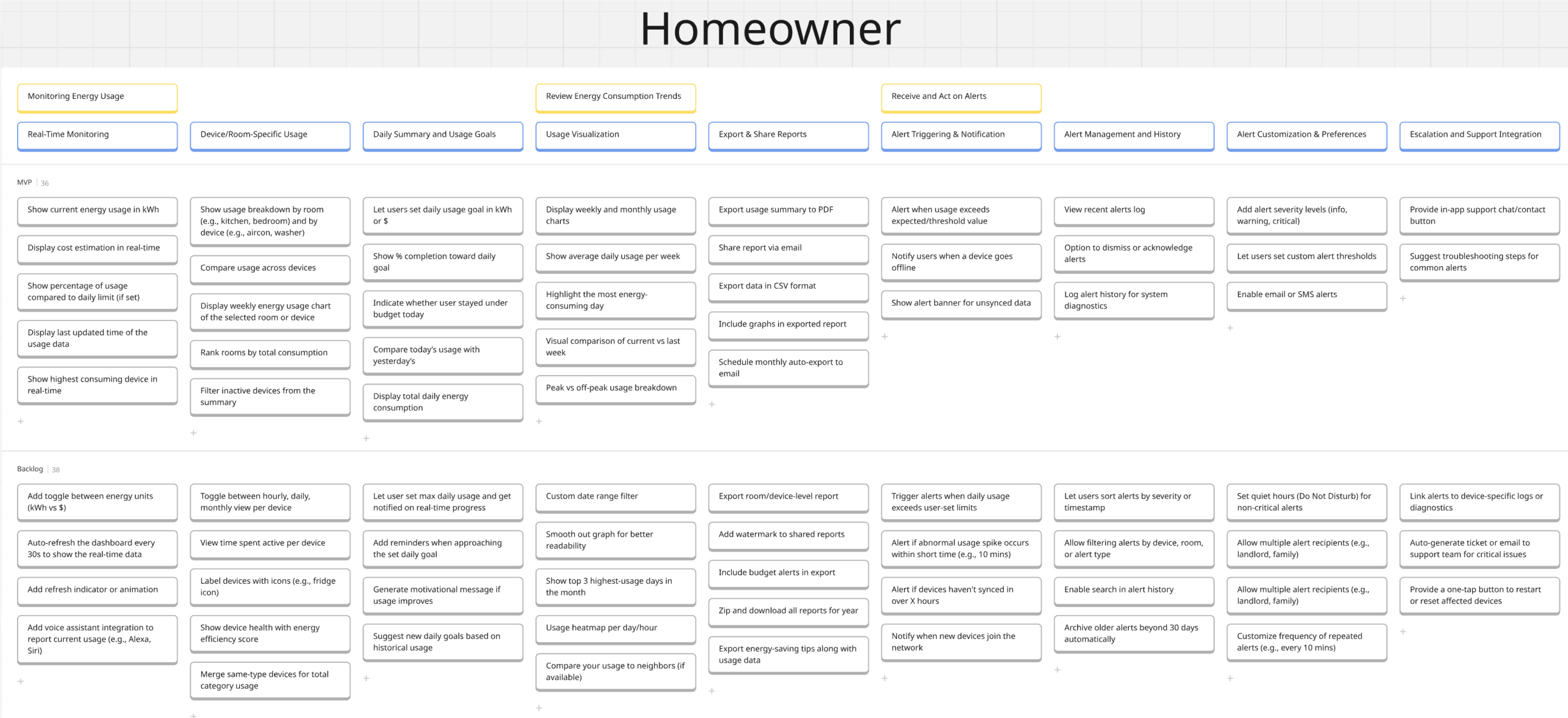
2.2 User Story Mapping

<u>User Personas</u>

1. Homeowner
2. System Administrator

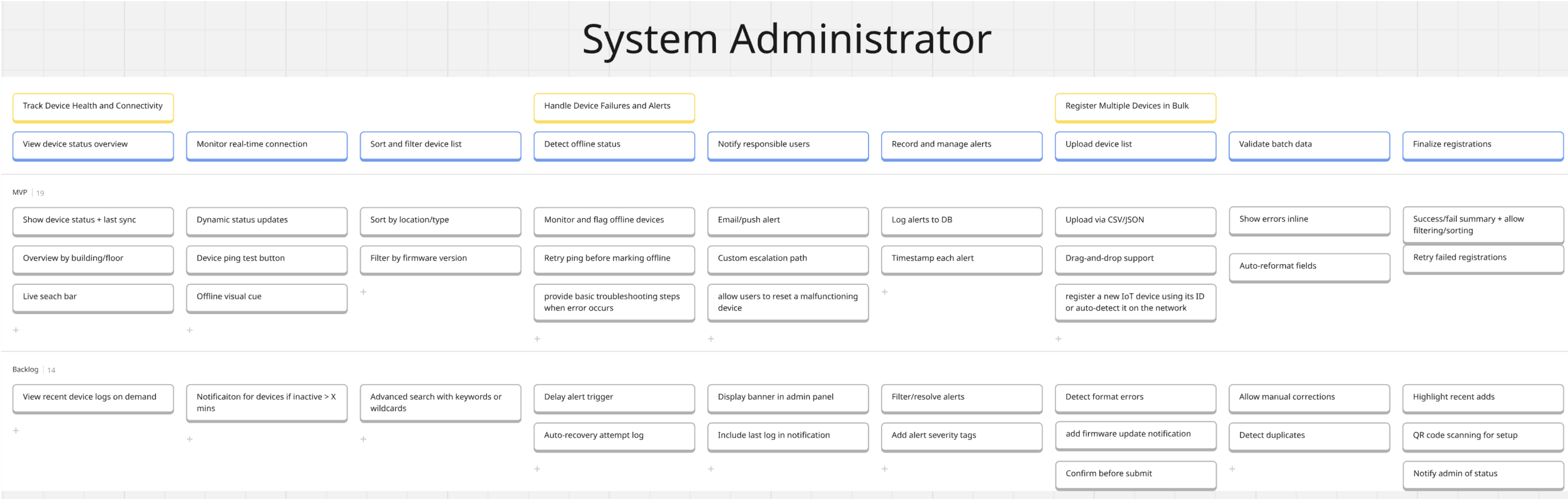
1. Homeowner

Set daily energy goal → Track real-time and room-specific usage → Review weekly/monthly trends → Receive alerts when usage exceeds limits → Take action or seek support



2. System Administrator

Register devices in bulk → Monitor device status → Receive offline or alert notifications → Log and escalate issues→ Resolve or retry registration/failure



3 Product Backlog

Backlog ID	Title	User story	Acceptance Criteria	Assigned to	Priority	Story Points
PBI-001	Enable homeowner to show current energy usage in kWh	As a homeowner, I want to view my current energy usage in kWh, so that I can track my real-time consumption and adjust usage immediately.	▪ The system should display current energy usage on the main dashboard in kWh. ▪ The usage data should refresh in near real-time (e.g., every 30 seconds). ▪ The displayed value should be accurate and match the backend data. ▪ The value should remain visible across sessions unless hidden by the user.	Zwe Htet Kyaw	4	5
PBI-011	Enable Homeowner to alert when usage exceeds expected/threshold value	As a homeowner, I want to control and monitor my energy usage based on a personalised threshold, so that I can be alerted when my usage exceeds the limit and ensure I stay within my budget.	▪ When I access the energy monitoring settings, then I should be able to set a threshold value (e.g., kWh per day/week/month), so that I can be notified if usage exceeds my expectation. ▪ When my energy usage goes above the defined threshold, then I should receive an alert with the severity level (e.g., warning or critical depending on how much it exceeds), ▪ When I receive the alert, then it should have my actual usage, the threshold I set, a possible explanation (e.g., which devices use more, peak hours) so that I understand what might be causing it ▪ When my energy needs change, then I should be able to update the threshold value or turn off the alert, so that it fits my evolving lifestyle.	Tan Pek Yu	4	5
PBI-021	Enable system administrator to monitor and flag offline devices	As a system administrator, I want to automatically monitor device connectivity and receive alerts when devices go offline, so that I can quickly identify and resolve connectivity issues before they impact operations.	▪ System automatically detects devices that haven't sent heartbeat signals within configurable timeframe (default 5 minutes) and flags them as offline ▪ Offline devices are visually distinguished in the device list with red status indicators and "OFFLINE" labels ▪ Dashboard displays total count of offline devices with last seen timestamps for each device ▪ Administrators can set custom offline detection thresholds per device type (1-60 minutes) ▪ System maintains offline device history log with duration tracking and reconnection timestamps	Tharun Ranjan	4	5
PBI-002	Enable homeowner to show usage breakdown by room/device	As a homeowner, I want to view energy usage categorized by individual rooms or devices, so that I can identify which appliances or areas are consuming the most electricity.	▪ The system should allow selection of specific rooms or devices. ▪ A breakdown of usage should be displayed for each selected room/device. ▪ The data should be presented visually (e.g., in bar chart or table format). ▪ Devices or rooms with zero usage should also be listed with 0 kWh.	Zwe Htet Kyaw	4	8
PBI-016	Enable System Administrator to View Device Status	As a system administrator, I want to view the online/offline status and last sync time of each device, so that I can monitor the network's health and detect inactive devices.	▪ The dashboard must show each registered device with its current online/offline status. ▪ The last sync time is displayed in a readable timestamp format. ▪ Status updates must occur within 10 seconds of a state change. ▪ Offline devices must be visually distinguished (e.g., greyed out or with a red dot).	Chan Ken Yew	4	5

Backlog ID	Title	User story	Acceptance Criteria	Assigned to	Priority	Story Points
			▪ Devices with no sync in the last 5 minutes should automatically be marked as offline.			
PBI-022	Enable system administrator to receive Email/push alert	As a system administrator, I want to receive email and push notifications when critical device events occur, so that I can respond promptly to issues even when not actively monitoring the dashboard.	▪ Administrators receive email notifications within 2 minutes of critical events (device offline, errors, threshold breaches) ▪ Push notifications are sent to mobile devices with customizable alert tones and priority levels ▪ Notification preferences can be configured per admin user (email only, push only, or both) ▪ Alert messages include device details, event type, timestamp, and direct links to device management page ▪ System tracks notification delivery status and provides retry mechanism for failed deliveries	Tharun Ranjan	3	8
PBI-013	Enable Homeowner to add alert severity levels (info/warning/critical)	As a homeowner, who doesn't have the time to read through all alerts provided, I want it to categorise alerts based on how important they are compared to others, so that I can have greater time efficiency as I don't have to scroll through all the alert	▪ When I receive an alert, then I should be able to clearly see whether it is informational, a warning, or critical, so I can understand how urgent it is ▪ When an alert is displayed, then it should be visually distinct based on severity (e.g., info = blue, warning = yellow, critical = red), so that I can quickly recognize how serious the issue is without reading the entire message. ▪ When I'm reviewing past alerts, then I should be able to filter or sort them by severity level, so that I can focus on critical issues first ▪ When I receive an alert, then it should have the correct severity level based on the situation, so that I am not overwhelmed by unnecessary panic or false urgency	Tan Pek Yu	4	5
PBI-012	Enable Homeowner to view recent alerts log	As a homeowner, I want to know how many times I have exceeded energy threshold in the past for a month etc., by having the recent alerts log, it allows me to keep track with how much I have exceeded my energy threshold, so that I can improve accordingly from these alerts	▪ When I click on the alert dashboard, I should see the alerts log ▪ The alerts log should show me a brief summary on past alerts ▪ When I click on the alert log, I should be able to see the past alerts in more details ▪ I should also be able to see past alerts according to how long ago it was, such as past week or past month ▪ By clicking on each past alert, I should be able to see in detail what caused the alert, how much energy exceeded threshold, which device it was	Tan Pek Yu	3	5
PBI-006	Enable Homeowner to view weekly/monthly usage charts	As a homeowner, I want to view a clear and visually engaging weekly energy usage chart, so that I can better understand my consumption trends, detect unusual spikes or drops, and take informed actions to reduce costs and improve efficiency in my household.	▪ The system should display a chart showing daily energy usage for the current week when the dashboard loads. ▪ The user should be able to select previous weeks, and the chart should update accordingly. ▪ Days with missing or incomplete data should still appear in the chart, displaying 0 kWh or an appropriate placeholder. ▪ When the user hovers over a data point, a tooltip should appear showing the exact energy usage for that day.	Choon Lin	4	8

Backlog ID	Title	User story	Acceptance Criteria	Assigned to	Priority	Story Points
PBI-007	Enable Homeowner to view the most energy- consuming day	As a homeowner, I want to view which day in the week had the highest energy consumption, so that I can better understand my peak usage patterns and adjust my habits to reduce energy waste and lower my utility bills.	▪ The system should display a **chart of energy usage** for each day of the **current week** when the energy dashboard loads. ▪ The user should be able to **select a previous week**, and the chart should update to reflect that week's data. ▪ If energy data is **missing or incomplete** for a day, the chart should **display 0 usage** or show an appropriate **placeholder**. ▪ When the user **hovers over a data point**, a **tooltip** should appear showing the **exact energy usage** for that day. ▪ The chart should **highlight or visually emphasize** the **most energy-consuming day** in the week to help users easily identify peak usage.	Choon Lin	3	5
PBI-008	Enable Homeowner to get the PDF Export of usage summary	As a homeowner, I want to export a detailed summary of my energy consumption as a downloadable PDF, so that I can archive my usage records, share them with consultants or household members, and monitor my progress in reducing energy consumption over time.	▪ When the user **clicks the "Export to PDF" button**, the system should **generate and automatically download a PDF** containing energy usage data for the selected week. ▪ The exported PDF should include the following elements: ▪ A **chart** showing **daily energy usage** – The **average daily usage** for the week – The **total weekly usage** – The **selected date range** – The **data and visuals** in the exported PDF should **match exactly** with what is displayed on the dashboard. ▪ If **no data is available** for the selected week, the system should **display an appropriate message**, such as: *"No data available to export."*	Choon Lin	4	8
PBI-020	Enable System Administrator to Notify Users When a Device Goes Offline	As a system administrator, I want the system to notify users when a device goes offline, so that they can take timely corrective actions.	▪ An offline notification is triggered within 30 seconds of a device disconnection. ▪ Notifications are sent via the configured method (e.g., push, email, or SMS). ▪ Users can acknowledge or dismiss the alert in the notification center. ▪ Offline alerts are stored in an alerts log for later review. ▪ No duplicate alerts are sent for the same device unless it goes online and offline again.	Chan Ken Yew	3	3
PBI-003	Enable homeowner to set daily usage goal in kWh or $	As a homeowner, I want to set a daily energy usage goal in either kWh or dollar value, so that I can track my consumption against a personal target and stay within budget.	▪ The system should allow users to input a daily energy goal in either kWh or dollars. ▪ The system should convert the value when toggling between kWh and $ if rates are available. ▪ The system should save the goal setting and retain it across sessions. ▪ The daily goal should be displayed clearly in the user dashboard.	Zwe Htet Kyaw	4	5
PBI-018	Enable System Administrator to Overview by Building/Floor	As a System Administrator, I want to view an aggregated overview of devices grouped by building and floor,	▪ Devices are grouped visually by building and floor. Each group shows the total number of devices and how many are online.	Chan Ken Yew	3	5

Backlog ID	Title	User story	Acceptance Criteria	Assigned to	Priority	Story Points
		so that I can monitor deployment health in physical locations.	▪ Admins can expand/collapse each building/floor group. ▪ Aggregates update in real time as device statuses change. ▪ Empty floors or buildings are hidden or marked appropriately.			
PBI-017	Enable System Administrator to Perform Device Sorting/Filtering	As a system administrator, I want to perform sorting and filtering on the list of devices by attributes such as location, type, and status, so that I can efficiently find and manage specific devices in large-scale environments.	▪ System Admins can sort devices by location, device type, and connection status. ▪ System Admins can filter devices using multiple criteria simultaneously (e.g., location + status). ▪ The device list updates dynamically based on the selected sort and filter options. ▪ The system retains the last used sort and filter preferences when the admin revisits the page. ▪ Filtered and sorted views return accurate results within acceptable performance limits.	Chan Ken Yew	3	3
PBI-005	Enable homeowner to add reminders when approaching the set daily goal	As a homeowner, I want to receive a notification when my energy usage is nearing my daily goal, so that I can take action early and avoid exceeding my intended usage limit.	▪ The system should send a soft reminder (e.g., pop-up or banner) when usage exceeds 80% of the set goal. ▪ A strong alert should be triggered when usage reaches or exceeds 100% of the goal. ▪ Users should be able to dismiss or snooze the reminder. ▪ The reminder should only appear once per session unless re-enabled.	Zwe Htet Kyaw	3	5
PBI-009	Enable Homeowner to get the room/device-level report	As a homeowner, I want to view a pie chart that breaks down energy usage by room or device, so that I can identify high-consumption areas and make informed decisions to reduce unnecessary energy usage.	▪ When the user is on the **energy usage page** and **device/room-level data is available**, a **pie chart** should be displayed showing the **percentage usage by room or device**. ▪ When the **user hovers over or clicks** on a pie chart section, the chart should **display the room/device name and exact energy used**. ▪ If **no detailed usage data exists**, the chart section should **show a message** like:*"Detailed usage data unavailable."* ▪ When the user **filters by a specific week or month**, the pie chart should **update to reflect energy usage by room/device for the selected period**.	Choon Lin	3	5
PBI-023	Enable system administrator to upload via CSV/JSON	As a system administrator, I want to bulk upload device configurations and data via CSV or JSON files, so that I can efficiently manage multiple devices without manual entry for each one.	▪ System accepts CSV and JSON files up to 10MB containing device configuration data and bulk operations ▪ File validation occurs on upload with clear error messages for format issues, missing required fields, or invalid data ▪ Upload progress bar shows real-time processing status with estimated completion time ▪ System provides detailed import summary showing successful records, failed records, and specific error descriptions ▪ Failed imports can be downloaded as error report files with corrected data templates for re-upload	Tharun Ranjan	4	

Backlog ID	Title	User story	Acceptance Criteria	Assigned to	Priority	Story Points
PBI-014	Enable Homeowner to set quiet hours	As a homeowner, I do not want to be constantly reminded about energy usage while I am going to bed and thus, a set quiet hours mode is nice, so that I don't receive active alerts at night.	▪ When I access the settings for my alert preferences, then I should be able to define a daily time range (e.g., 10:00 PM to 7:00 AM) during which alerts are muted or limited. ▪ When quiet hours are active and an alert is triggered, then the alert should still be logged and visible in the app, so that I can review anything I missed once quiet hours end ▪ When I want to receive alerts during quiet hours, then I should have the option to temporarily disable quiet mode ▪ When a critical alert is triggered during quiet hours, then I should still receive the notification immediately.	Tan Pek Yu	2	3
PBI-019	Enable System Administrator to View Recent Logs on Demand	As a System Administrator, I want to view the recent activity logs of a selected device on demand, so that I can quickly investigate behavior or issues.	▪ Admins can open a panel to view logs for a selected device. ▪ The log view shows timestamped entries from the past 24–48 hours. ▪ Logs are pulled only when requested to save performance. ▪ A message is shown if no logs are available. ▪ Logs are shown in reverse chronological order.	Chan Ken Yew	2	3
PBI-010	Enable Homeowner to do custom date range filter	As a homeowner, I want to compare energy usage across multiple weeks or months, so that I can spot trends, evaluate progress, and adjust my habits more effectively over time.	▪ When the user is on the **energy dashboard** and selects **two or more weeks or months**, the system should display a **side-by-side or overlaid comparison chart** of the selected timeframes. ▪ When the **comparison data is displayed**, hovering over any data point should **reveal the exact energy value and the corresponding date range**. ▪ If the user selects a **time range with no data**, the system should display a **message** such as: *"No data available for this time range."* ▪ When the user selects a **new time for comparison**, the chart should **refresh automatically with updated comparison data**.	Choon Lin	2	3
PBI-004	Enable homeowner to add toggle between energy units (kWh vs $)	As a homeowner, I want to toggle between viewing my energy usage in kWh and dollars, so that I can better understand both my consumption and its financial impact.	▪ The system should offer a toggle button to switch between kWh and dollar view. ▪ When toggled, all usage metrics should update accordingly. ▪ The conversion should be based on a predefined rate stored in the system. ▪ The system should revert back to the last selected unit on reload.	Zwe Htet Kyaw	3	3
PBI-025	Enable system administrator to confirm before submission	As a system administrator, I want confirmation dialogs before executing critical actions, so that I can prevent accidental changes that could disrupt device operations.	▪ Confirmation dialogs appear for all destructive actions including device deletion, factory reset, and bulk operations ▪ Confirmation modals display affected items count, action type, and potential impact warnings ▪ Users must explicitly click "Confirm" or type confirmation text for high-risk actions like bulk deletion ▪ Confirmation dialogs show device names/IDs and operation details for review before proceeding	Tharun Ranjan	3	5

Backlog ID	Title	User story	Acceptance Criteria	Assigned to	Priority	Story Points
			▪ System provides "Cancel" option that returns user to previous state without executing any changes			
PBI-024	Enable system administrator to delay alert trigger	As a system administrator, I want to configure delay periods before alerts are triggered, so that I can avoid false alarms from temporary network glitches or brief device maintenance.	▪ Configurable delay periods (1-60 minutes) can be set per alert type to prevent false alarms from temporary issues ▪ Countdown timers are displayed in the UI showing remaining time before alert activation ▪ Alerts are automatically cancelled if triggering conditions resolve during the delay period ▪ System logs show delay period settings and whether alerts were cancelled or proceeded after delay ▪ Different delay settings can be applied to different device types and alert severity levels	Tharun Ranjan	2	3
PBI-015	Enable Homeowner to show alert banner for unsynced data	As a homeowner, I want to be notified when my energy data is not properly synced, so that I can ensure the alerts I receive are based on accurate and up-to-date information, and take appropriate action when necessary.	▪ When I log in, data should be synced accordingly ▪ If the data isn't in sync, my system should send me a notification regarding the unsynced data ▪ The notification I receive is classified as important ▪ When I click on the notification, I should be directed directly to the alert's dashboard ▪ In the alert's dashboard, I should see the alert highlighted and more details regarding the issue ▪ I should also see ways to fix the issue	Tan Pek Yu	2	3

4 Sprint Backlog with Definition of Done

4.1 Sprint Goal

Sprint-1*:* Establish core monitoring functionality for homeowners and ensure system administrators are alerted about critical device issues.

Sprint-2: Expand user visibility into consumption patterns and ensure complete administrative device onboarding and logging.

Sprint-3: Finalize reporting, customization, and alert management features for both personas.

4.2 Definition of Done for Sprint 1

- Features for both Homeowner and System Administrator are working properly and linked to real-time usage or device data
- All acceptance criteria listed in each user story are fulfilled and checked by team members
- UI is tested on both desktop and mobile screens to make sure everything looks consistent and usable

- Code is reviewed by at least one other teammate and functions without major errors
- Homeowner features (like usage tracking and alerts) are tested using sample energy values to check updates and thresholds
- System Administrator features (like device status and offline detection) are tested using mock device data to ensure accurate results
- Feedback is collected from within the team to improve clarity and design
- No critical bugs or blockers during walkthrough and test cases

4.3 User Stories and Tasks for Sprint 1

User Story ID	Title	User Stories	Assigned To	Priority	Task(s)	Estimation (in hours)
PBI-001	Enable homeowner to show current energy usage in kWh	As a homeowner, I want to view my current energy usage in kWh, so that I can track my real-time consumption and adjust usage immediately.	Zwe Htet Kyaw	4	▪ Design UI mockup for energy usage card (1.5 h) ▪ Set up backend API to fetch real-time usage data (2 h) ▪ Integrate API with front-end component (2.5 h) ▪ Test real-time data refresh and value accuracy (1.5 h) ▪ Update documentation/user guide (0.5 h)	8
PBI-002	Enable homeowner to show usage breakdown by room/device	As a homeowner, I want to view energy usage categorized by individual rooms or devices, so that I can identify which appliances or areas are consuming the most electricity...	Zwe Htet Kyaw	4	▪ Design layout for room/device breakdown UI (1.5 h) ▪ Update database or backend logic to group data (2 h) ▪ Build frontend logic to display grouped usage (2.5 h) ▪ Implement filtering/sorting interactions (2 h) ▪ Unit + UI testing for multiple rooms/devices (1 h)	9
PBI-006	Enable Homeowner to view weekly/monthly usage charts	As a homeowner, I want to view a clear and visually engaging weekly energy usage chart, so that I can better understand my consumption trends, detect unusual spikes or drop and take informed actions to reduce costs and improve efficiency in my household.	Choon Lin	4	▪ The system should display a chart showing daily energy usage for the current week when the dashboard loads ▪ The user should be able to select previous weeks, and the chart should update accordingly. ▪ Days with missing or incomplete data should still appear in the chart, displaying 0 kWh or an appropriate placeholder. ▪ When the user hovers over a data point, a tooltip should appear showing the exact energy usage for that day.	9
PBI-011	Enable Homeowner to alert when usage exceeds expected/threshold value	As a homeowner, I want to control and monitor my energy usage based on a personalised threshold, so that I can be alerted when my usage exceeds the limit and ensure I stay within my budget.	Tan Pek Yu	4	▪ Design threshold-setting interface for homeowners (UI input/slider) (1.5 hr) ▪ Develop alert notification system (in-app and/or email/push) (2 hr) ▪ Implement backend logic to evaluate real-time usage vs. Threshold (2.5 hr) ▪ Store and manage user-defined threshold values in database (1 hr)	8

User Story ID	Title	User Stories	Assigned To	Priority	Task(s)	Estimation (in hours)
					▪ Test threshold alert scenarios (e.g., exact match, over-limit, reset) (1 hr)	
PBI-016	Enable System Admins to View Device Status	As a system administrator, I want to view the online/offline status and last sync time of each device, so that I can monitor the network's health and detect inactive devices.	Chan Ken Yew	4	▪ Develop backend status tracking logic (4h) ▪ Design status and sync UI layout (2h) ▪ Integrate UI with live status data (2h)	8
PBI-017	Enable System Admins to Perform Device Sorting/Filtering	As a system administrator, I want to perform sorting and filtering on the list of devices by attributes such as location, type, and status, so that I can efficiently find and manage specific devices in large-scale environments.	Chan Ken Yew	3	▪ Design sorting and filtering UI components (2h) ▪ Implement backend filtering and sorting logic (2.5h) ▪ Integrate and test filter/sort functionality (1.5h)	6
PBI-021	Enable system administrator to monitor and flag offline devices	As a system administrator, I want to automatically monitor device connectivity and receive alerts when devices go offline, so that I can quickly identify and resolve connectivity issues before they impact operations.	Tharun Ranjan	4	▪ Design UI for device status dashboard with offline indicators (1 h) ▪ Set up backend monitoring service to track device heartbeats (2.5 h) ▪ Implement offline detection logic with configurable thresholds (1.5 h) ▪ Create alert system for offline device notifications (1.5 h) ▪ Build offline device filtering in device list (1 h) ▪ Unit and integration testing for monitoring service (0.5 h)	8
PBI-022	Enable system administrator to receive Email/push alert	As a system administrator, I want to receive email and push notifications when critical device events occur, so that I can respond promptly to issues even when not actively monitoring the dashboard.	Tharun Ranjan	3	▪ Design notification preferences UI for admin settings (1 h) ▪ Set up email service integration (SMTP/SendGrid) (2.5 h) ▪ Implement push notification service (Firebase/OneSignal) (2.5 h) ▪ Create notification templates for different alert types (1.5 h) ▪ Build notification queuing and delivery logic (1 h) ▪ Test email and push notification reliability (0.5 h)	9

Total hours for Sprint 1 = 65 hours

Total no of PBIs = 8

PDF of Sprint Plan for all 3 sprints

https://1drv.ms/b/c/cff61086a24dec14/EfI7-5kp62pOs0aogyJCu2MBkOoD-6BdAgvTFUWZDAhXuA?e=TJjLbl
(some browsers cannot open the OneDrive link, so we included the pdf just in case)

Sprint 1- User Stories Allocation for each mem

Zwe Htet Kyaw - Monitoring Energy Usage (Homeowner)

- MVP - Show current energy usage in kWh (PBI-001)
- MVP - Show usage breakdown by room/device (PBI-002)
- MVP - Let users set daily usage goal in kWh or $ (PBI-003)
- Backlog - Add toggle between energy units (kWh vs $) (PBI-004)
- Backlog - Add reminders when approaching the set daily goal (PBI-005)

Choon Lin - Review Energy Consumption Trends (Homeowner)

- MVP - Display weekly and monthly usage charts (PBI-006)
- MVP - Highlight most energy-consuming day (PBI-007)
- MVP - Export usage summary to PDF (PBI-008)
- Backlog - Custom date range filter (PBI-009)
- Backlog - Export room/device-level report (PBI-010)

Tan Pekyu - Receive and Act on Alerts (Homeowner)

- MVP - Alert when usage exceeds expected/threshold value (PBI-011)
- MVP - View recent alerts log (PBI-012)
- MVP - Add alert severity levels (info, warning, critical) (PBI-013)
- Backlog - Set quiet hours (do not disturb mode) (PBI-014)
- Backlog - Link alerts to device-specific logs (PBI-015)

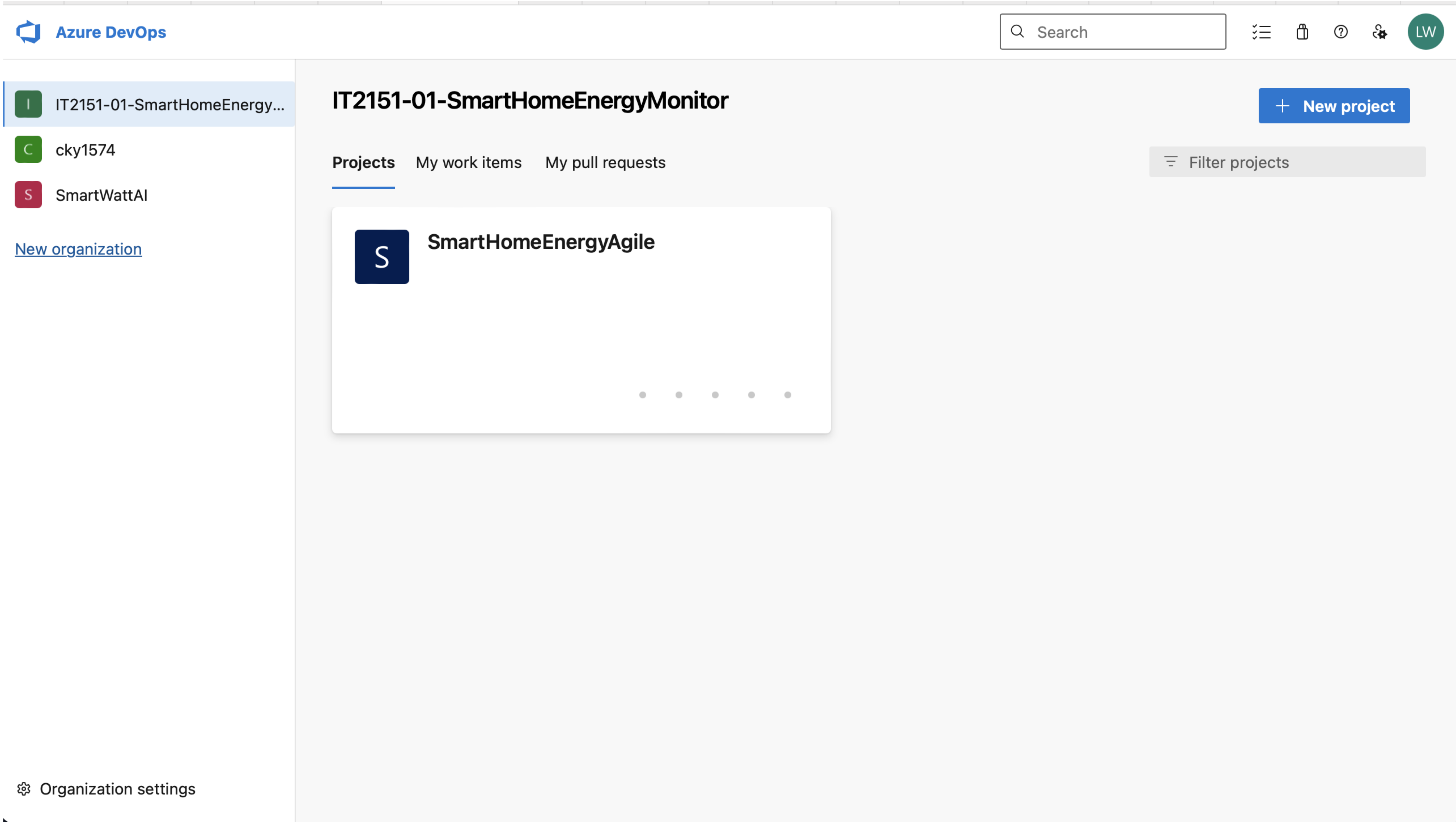
Chan Ken Yew - Track Device Health and Connectivity (System Administrator)

- MVP - Show device status + last sync (PBI-016)
- MVP - Sort by location/type (PBI-017)
- MVP - Overview by building/floor (PBI-018)
- Backlog - View recent device logs on demand (PBI-019)
- Backlog - Notification for offline devices > x mins (PBI-020)

Tharun Ranjan - Handle Device Failures and Alerts, Register Multiple Devices in Bulk (System Administrator)

- MVP - Monitor and flag offline devices (PBI-021)
- MVP - Email/push alert to admins (PBI-022)
- MVP - Upload via CSV/JSON (PBI-023)
- Backlog - Delay alert trigger (PBI-024)
- Backlog - Confirm before submission (PBI-025)

Our Azure DevOps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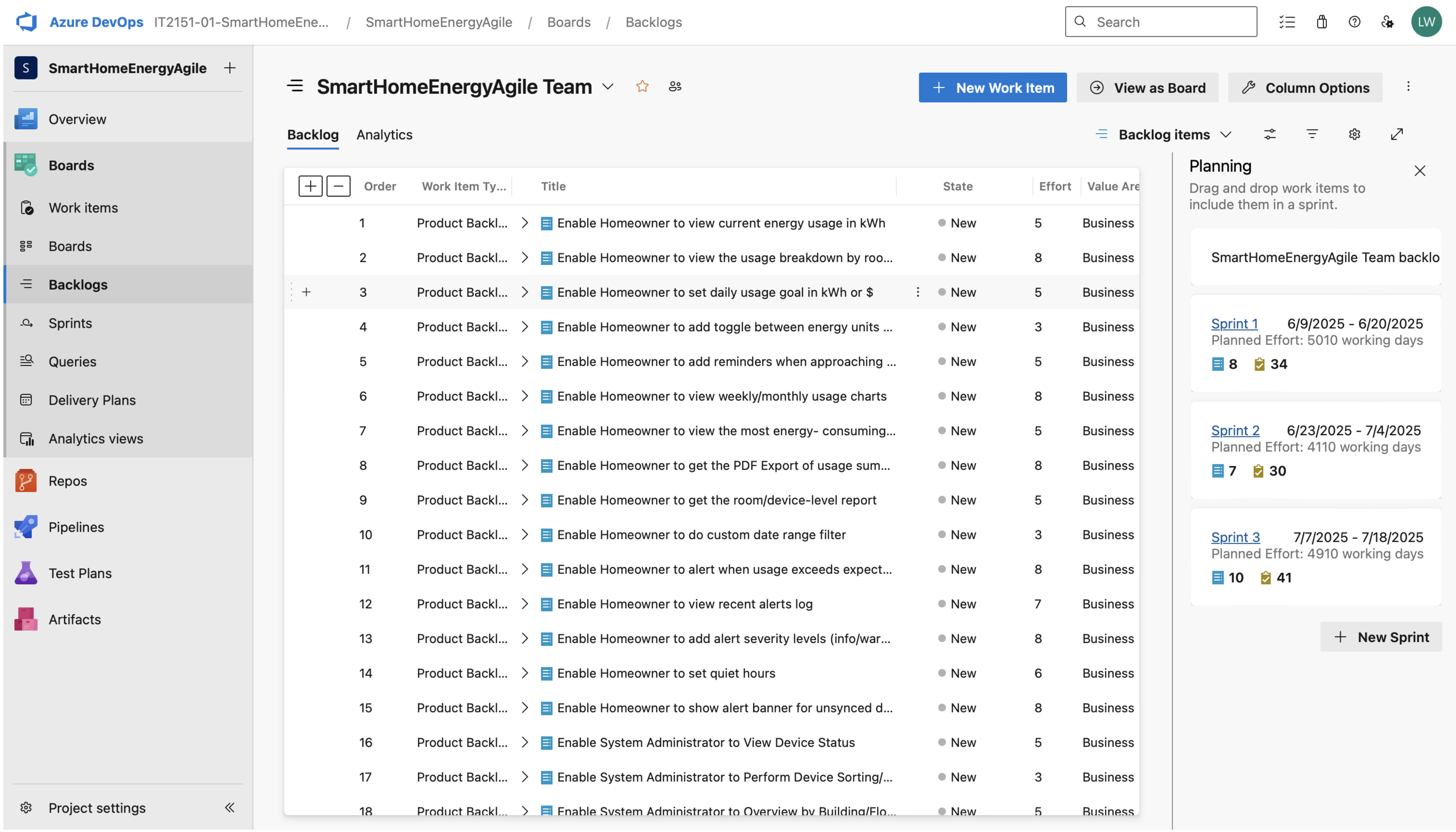
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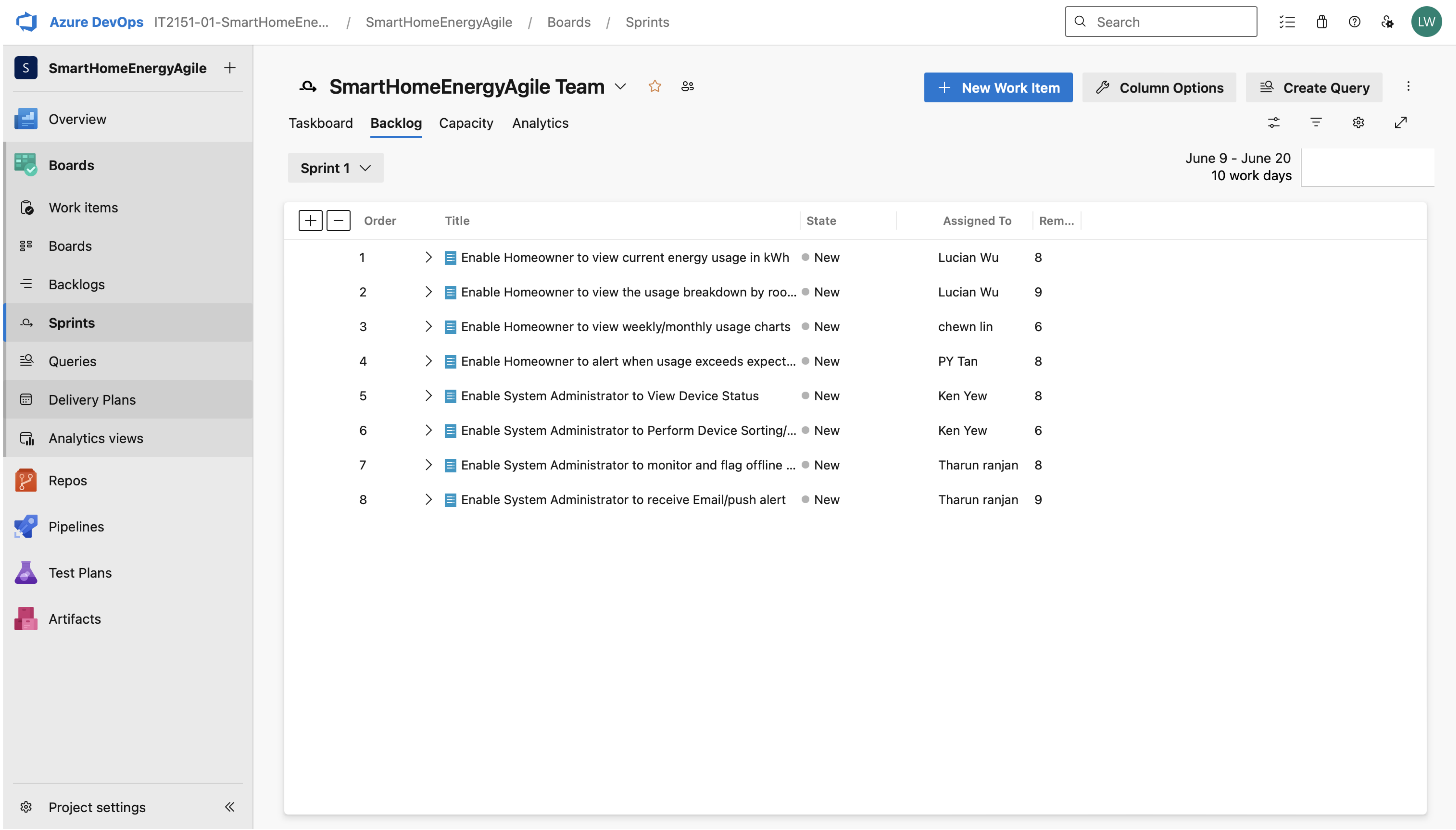
1 Product Backl... Enable Homeowner to view current energy usage in kWh New 5 Business
2 Product Backl... Enable Homeowner to view the usage breakdown by roo... New 8 Business
3 Product Backl... Enable Homeowner to set daily usage goal in kWh or $ New 5 Business
4 Product Backl... Enable Homeowner to add toggle between energy units ... New 3 Business
5 Product Backl... Enable Homeowner to add reminders when approaching ... New 5 Business
6 Product Backl... Enable Homeowner to view weekly/monthly usage charts New 8 Business
7 Product Backl... Enable Homeowner to view the most energy- consuming... New 5 Business
8 Product Backl... Enable Homeowner to get the PDF Export of usage sum... New 8 Business
9 Product Backl... Enable Homeowner to get the room/device-level report New 5 Business
10 Product Backl... Enable Homeowner to do custom date range filter New 3 Business
11 Product Backl... Enable Homeowner to alert when usage exceeds expect... New 8 Business
12 Product Backl... Enable Homeowner to view recent alerts log New 7 Business
13 Product Backl... Enable Homeowner to add alert severity levels (info/war... New 8 Business
14 Product Backl... Enable Homeowner to set quiet hours New 6 Business
15 Product Backl... Enable Homeowner to show alert banner for unsynced d... New 8 Business
16 Product Backl... Enable System Administrator to View Device Status New 5 Business
17 Product Backl... Enable System Administrator to Perform Device Sorting/... New 3 Business
18 Product Backl... Enable System Administrator to Overview by Building/Flo... New 5 Business
Planning
Drag and drop work items to include them in a sprint.
SmartHomeEnergyAgile Team backlo
Sprint 1 6/9/2025 - 6/20/2025
Planned Effort: 5010 working days
8 34
Sprint 2 6/23/2025 - 7/4/2025
Planned Effort: 4110 working days
7 30
Sprint 3 7/7/2025 - 7/18/2025
Planned Effort: 4910 working days
10 41
New Sprint


Azure DevOps IT2151-01-SmartHomeEne... / SmartHomeEnergyAgile / Boards / Sprints
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June 9 - June 20
10 work days
Order Title State Assigned To Rem...
1 Enable Homeowner to view current energy usage in kWh New Lucian Wu 8
2 Enable Homeowner to view the usage breakdown by roo... New Lucian Wu 9
3 Enable Homeowner to view weekly/monthly usage charts New chewn lin 6
4 Enable Homeowner to alert when usage exceeds expect... New PY Tan 8
5 Enable System Administrator to View Device Status New Ken Yew 8
6 Enable System Administrator to Perform Device Sorting/... New Ken Yew 6
7 Enable System Administrator to monitor and flag offline ... New Tharun ranjan 8
8 Enable System Administrator to receive Email/push alert New Tharun ranjan 9

Acknowledgements and References

We would like to acknowledge the use of several resources that have guided our understanding and development of Agile practices for the *Smart Home Energy Monitor* project.

1. **Lab Exercises and Lecture Notes**

   Much of the foundational knowledge for writing user stories, creating PBIs (Product Backlog Items), and planning sprints was derived from our Wk 5, 6 and 7 coursework materials. These resources helped us apply theoretical concepts in a hands-on context.

2. **Online Reference for Agile Concepts**

   For further theoretical clarification on Agile methodologies, user stories, PBIs, and sprint planning, we referred to:

   Atlassian Agile Coach: User Stories | Atlassian
   https://www.atlassian.com/agile/project-management/epics-stories-themes
   This resource provided structured guidance on writing effective user stories, estimating work, and managing backlogs within Agile teams.

3. **Use of AI Tools (ChatGPT)**

   We responsibly utilized ChatGPT by OpenAI to support our learning process. The AI tool was used only to supplement our studies by helping us explore unclear concepts, refine our user stories and tasks, and gather explanations for theoretical ideas. Every final work in the project was done by us based on our understanding of the Agile framework.